

# Robotics Math - Assignment 01

> Suraj Dayma | U20230102

## Task 1

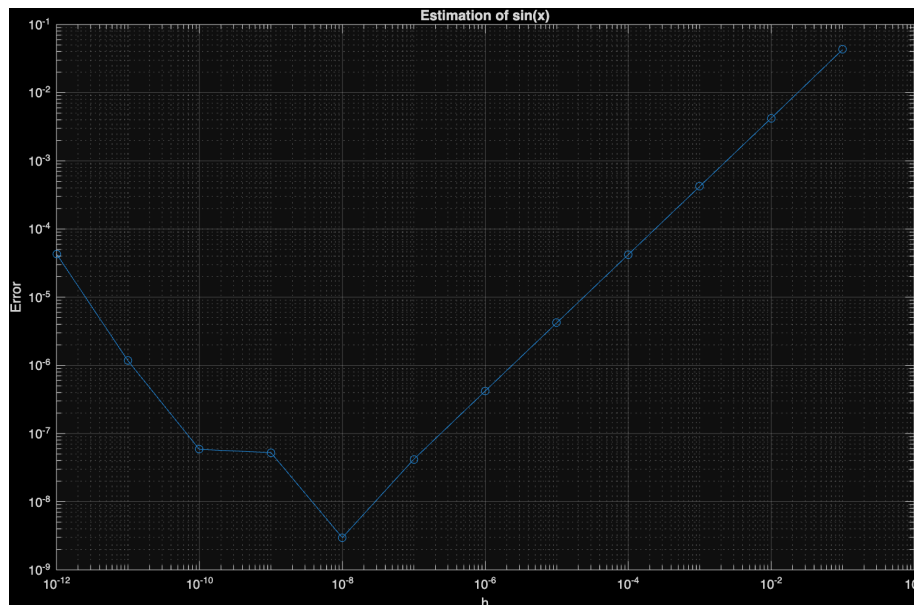
Master all parts of Prata MATLAB book.

- I have done a little MATLAB before, so I went straight into doing the rest of the tasks.

## Task 2

1. Use the definition of derivative to calculate the derivative of  $\sin(x)$  at  $x = 1$ .
2. Compare that to the exact value (meaning  $\cos(1)$ ).
3. Plot error vs  $h$  (log log plot) and note that there is a best  $h$  because of the competition between method error and roundoff error.
4. Extra credit, using the Matlab resolution of Matlab's "eps" (look it up), estimate the best  $h$  for this calculation and compare it with the minimum on your graph.

The code for this task can be found in [`sin\_error.m`](#)



The results for error vs  $h$  on a loglog plot.

We can see that after we get the lowest error at  $h_{\text{best}} = 10^{-8}$ .

When  $h > h_{\text{best}}$  we observe more method error.

When  $h < h_{\text{best}}$  we observe more roundoff error.

Task 2.4:

$$\frac{d}{dx} \sin(x) = \cos x + E = \frac{\sin(x+h) - \sin(x)}{h}$$

Method  $E = \frac{\sin(x+h) - \sin(x)}{h} - \cos x$

$$\sin(x+h) = \sin(x) + h \cos x - \frac{h^2}{2} \sin x$$

$$\frac{\sin(x+h) - \sin(x)}{h} - \cos x = \frac{h}{2} \sin x$$

$\Rightarrow E$  is on order of  $h$

MATLAB  $\epsilon$  : smallest difference between two floating point numbers.

$$\Rightarrow \sin(x+h) - \sin(x) \geq \epsilon$$

$$h \cos x \geq \epsilon$$

$$\frac{d}{dx} \sin x = \cos x \geq \frac{\epsilon}{h} = \text{Roundoff error.}$$

$$E_{\text{total}} = h + \frac{\epsilon}{h}$$

$$\frac{d}{dh} E_t = 1 - \frac{1}{h^2} \epsilon$$

$$\text{Set } \frac{d}{dh} E_t = 0 \Rightarrow h = \sqrt{\epsilon}$$

$$\text{IN MATLAB, } \epsilon = \epsilon \approx 10^{-16}$$

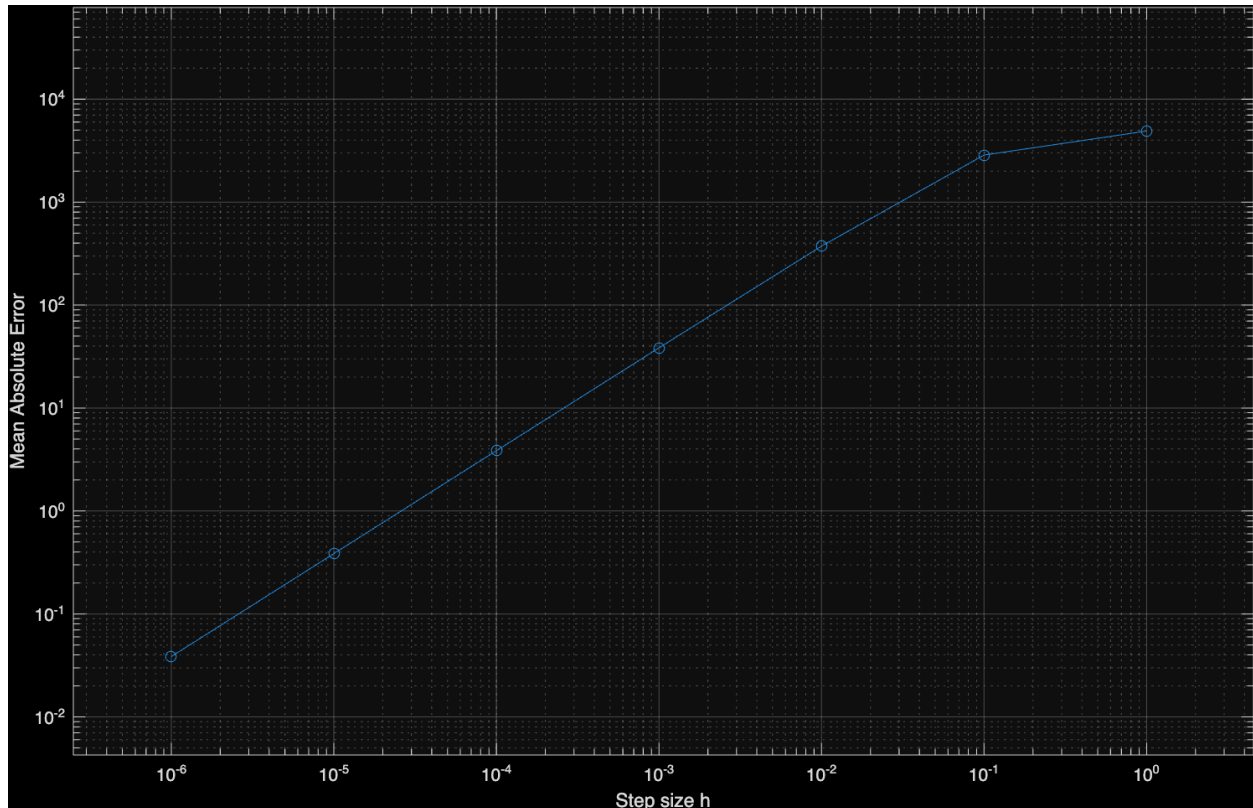
$$\therefore \underline{h_{\text{best}}} = 10^{-8}$$

( Same as  $h$  vs error plot.

## Task 3, 4, 5

- 3) Complete the exercises from the Tutorial on Thursday morning 29 January.
- 4) Demonstrate that you can write code equivalent to what I have attached here. If you do that first, you can use it to make the error plots asked for in class Thursday 29 Jan. You should be able to write it using Matlab help and no more.

Done in file *ErrorAnalysis.m*



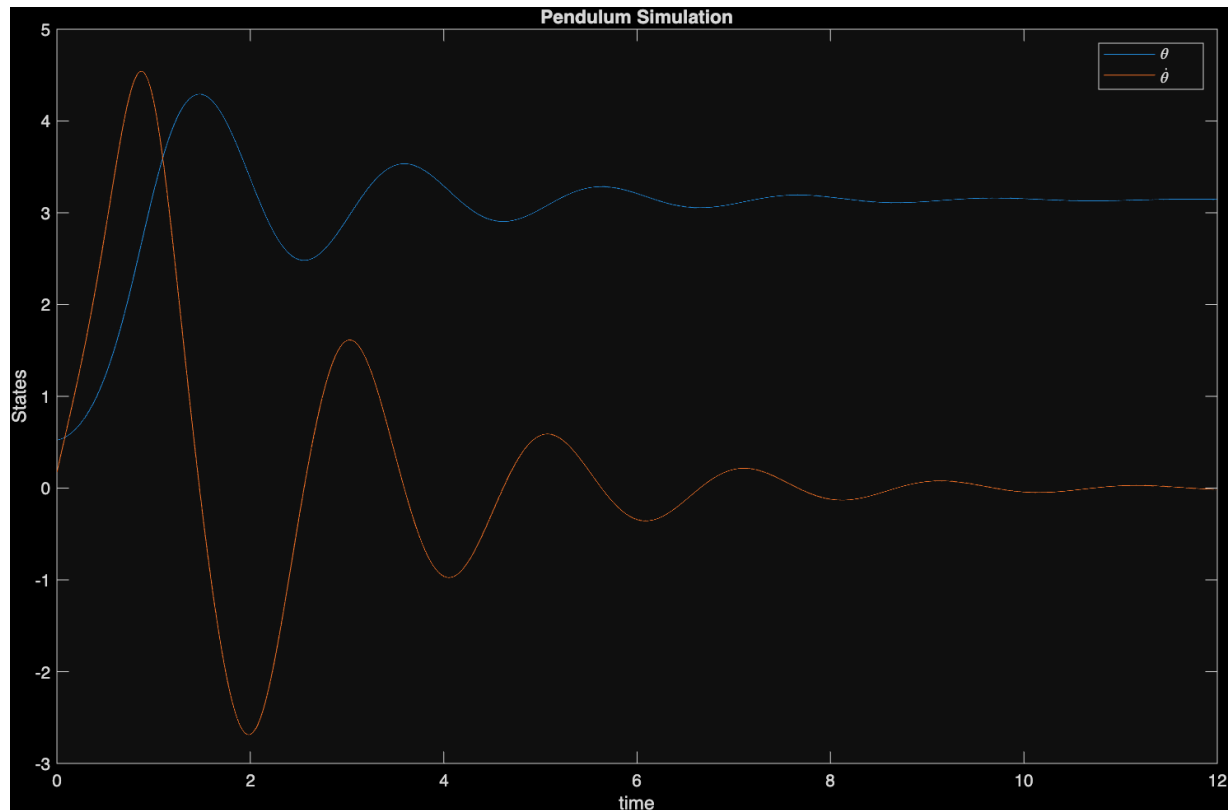
I was not able to compute the error for more points. This is due to the following reason:

- In the euler solver function, I initialize an array of n points.
- When  $n = 10^9$ , that would require  $10^9 * 8 \text{ bytes} = 8 \text{ GB}$  of memory for a single array.
- I had an array for t, z, and z\_exact  $\Rightarrow 8 * 3 = 24 \text{ GB}$  of memory for the entire computation.
- MATLAB gave me a warning about this and stopped my program.
- This could be mitigated by computing the error every iteration of the Euler loop and computing a rolling average for every value h, but I didn't bother with this.

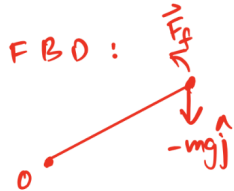
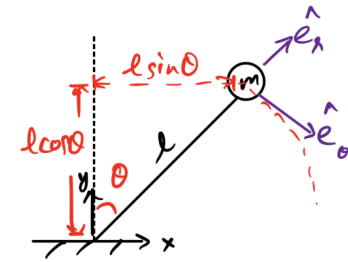
5) Cook up, or find, some ODE, or set of first order ODEs, and solve them and explore something hopefully-interesting about the solution.

Done in file *interestingODE.m*

The equations of motion of an inverted pendulum with air resistance were derived as shown below and the dynamics were simulated using Euler's method.



The dynamics of this is derived below.



$$\vec{r} = l \hat{e}_r, \quad \vec{v} = l \dot{\theta} \hat{e}_\theta$$

$$\vec{a} = l \ddot{\theta} \hat{e}_\theta - l \dot{\theta}^2 \hat{e}_r$$

$$\text{Air Friction } \vec{F}_f = -\gamma l \dot{\theta} \hat{v} = -\gamma l \dot{\theta} \hat{e}_\theta$$

Angular momentum balance,

$$\{ + \sum \vec{M}_O = \vec{r} \times m \vec{a} \} \cdot \hat{k}$$

$$\Rightarrow (\vec{r} \times \vec{w}) + \vec{r} \times (-\gamma l \dot{\theta}) \hat{e}_\theta = \vec{r} \times m(l \ddot{\theta} \hat{e}_\theta - l \dot{\theta}^2 \hat{e}_r)$$

$$\Rightarrow -\gamma l^2 \dot{\theta} + l m g \sin \theta = m l^2 \ddot{\theta}$$

states :  $\theta, \dot{\theta}$

$$\dot{z}_1 = \dot{\theta} = z_2 = \dot{\theta}$$

$$\dot{z}_2 = \ddot{\theta} = \frac{1}{m l^2} \left( \underbrace{-\gamma l^2 z_2}_{\dot{\theta}} + l m g \sin(z_1) \right)$$